$SVM_{Dynamic}$ — An Suite of Learning, Indexing and Sampling Algorithms for Grasping Complex Query Concepts

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Abstract

Specifying exact query concepts has become increasingly challenging to end-users. This is because many query concepts (e.g., those for looking up a multimedia object or seeking for a solution to a particular problem) can be hard to articulate, and articulation can be subjective. This invention proposes a suite of algorithms that learns query criteria through an intelligent sampling process. This suite of algorithms, which we call $SVM_{Dynamic}$ consists of an active learning algorithm, a sampling algorithm, and a high-dimensional indexing algorithm. Individually, these algorithms are novel by themselves. Together, they can model very complex query concepts and learn a target concept both quickly and a minimum number of samples.

$SVM_{Dynamic}$ can make a marked difference for e-commerce applications, on-line search engines, and any Internet sites that concern customer care, since it learns what an typically impatient user's need through a quick and intelligent Q&A session. Through examples, analysis and experiments, we have proven that $SVM_{Dynamic}$ converges to query concepts significantly faster than traditional methods.

Keywords: query concept, relevance feedback, active learning, high-dimensional indexing, sampling.

1 Introduction

The query-concept learning approach that we research in this study can be characterized by the following example: Suppose one is asked, "Are the paintings of Leonardo da Vinci more like those of Peter Paul Rubens or those of Raphael?" One is likely to respond with: "What is the basis for the comparison?" Indeed, without knowing the criteria (i.e., the query concept) by which the comparison is to be made, a database system cannot conduct a search. For many search tasks, however, a query concept is hard to articulate, and articulation can be subjective. For instance, in a multimedia search, it is difficult to describe a desired image using low-level features such as color, shape and texture (these are widely used features for representing images). Different users may use different combinations of these features to depict the same image. In addition, most users (e.g., Internet users) are not trained

to specify simple query criteria using SQL, for instance. In order to take individuals' subjectivity into consideration and to make information access easier, it is both necessary and desirable to build intelligent search engines that can discover (i.e., that can learn) individuals' query concepts quickly and accurately.

To learn users' query concepts, we propose a query-concept learner that learns a concept through an intelligent sampling process. The design of a query-concept learner must fulfill two primary goals. One, the concept-learner's hypothesis space must not be too restrictive, so it can model most practical query concepts. Two, the concept-learner should grasp a concept quickly and with a small number of labeled instances, since most users do not wait around to provide a great deal of feedback. To fulfill these design goals, we propose $SVM_{Dynamic}$, which consists of three novel algorithms:

1. An active learning algorithm.
2. An efficient sampling algorithm.
3. An high-dimensional indexing algorithm.

Our evaluation shows that $SVM_{Dynamic}$ learns query concepts substantially faster than traditional schemes. Our experiments with a high-dimensional image dataset confirms the effectiveness of $SVM_{Dynamic}$ in learning complex query concepts both accurately and quickly. We believe that a scheme like $SVM_{Dynamic}$, which can model most query concepts and can learn them quickly can make information access easier, faster, and more personal and user-friendly.